

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

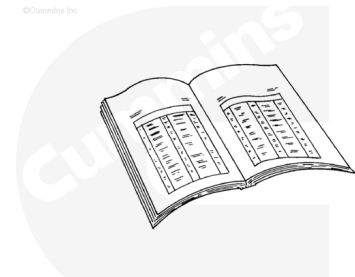
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Rotating the engine with the connecting rod caps loose or removed can cause damage to the piston cooling nozzles. Always remove the piston cooling nozzles prior to removing the connecting rod and piston assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-025-tr.html>)
- Remove the oil transfer connection. Refer to Procedure 007-061 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-061-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-035-tr.html>)

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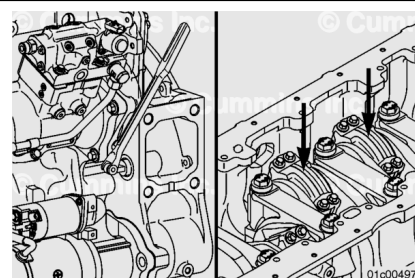


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- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-089-tr.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-046-tr.html)

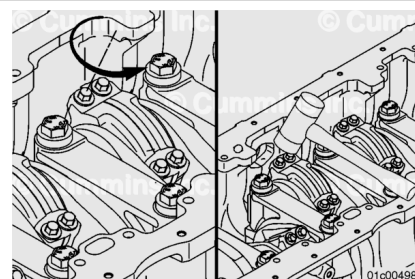
Remove

Rotate the crankshaft so that two of the connecting rods are at bottom dead center.



⚠ CAUTION ⚠

Do not damage the fractured split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fractured split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the rod.



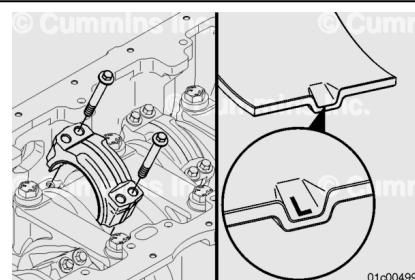
Do **not** remove the capscrews at this time.

Loosen the connecting rod capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the rod caps from the dowels.

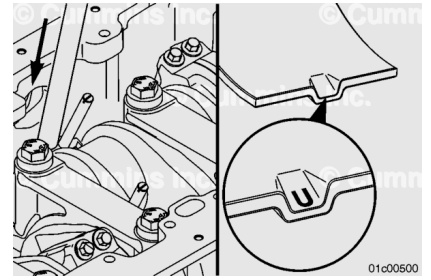
Remove the capscrews and the rod cap.

Remove the bearing shell from the rod cap and mark the cylinder number and the letter "L" on the flat surface of the bearing tangs.



Use a T-handle piston pusher to push the rod up far enough to allow the upper bearing shell to be removed.

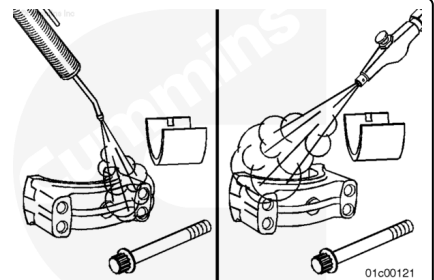
Remove the bearing shell and mark the cylinder number and the letter "U" on the flat surface of the bearing tangs.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

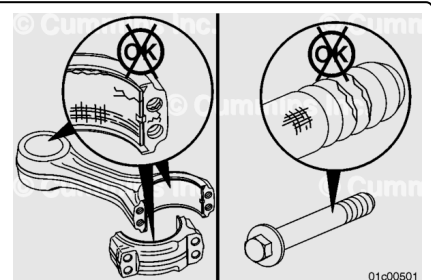
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the parts and dry with compressed air.

Inspect the rod caps, connecting rod bearing saddles, and capscrews for nicks, cracks, burrs, scratches, and fretting.

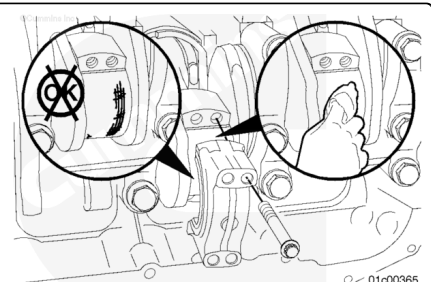
Polished spots on either the connecting rod or connecting rod cap indicate fretting and indicate that the rod **must** be replaced.

Dark areas on the face indicate carbonized oil deposits. Connecting rods with dark areas are reusable, if fretting is **not** present.



Inspect the connecting rod journal of the crankshaft for scratches or nicks.

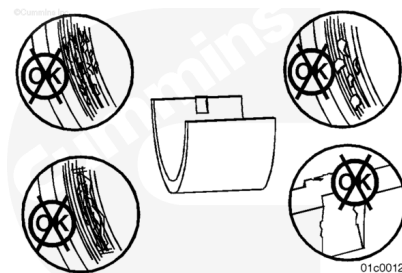
Use crocus cloth to remove the nicks and scratches.



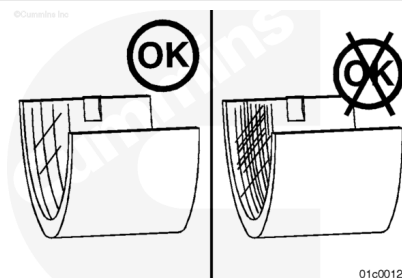
Inspect the bearings for damage.

Replace any bearings with the following damage:

- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches (deep enough to be felt with a fingernail).



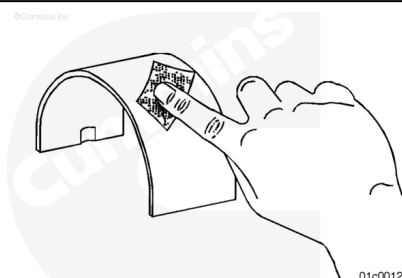
Normal bearing wear produces a smooth finish that will wear into the lining. Refer to the Parts Reuse Guidelines, Bulletin 3810303.



For more detailed information of bearing damage, refer to Analysis and Prevention of Bearing Failures, Bulletin 3810387.

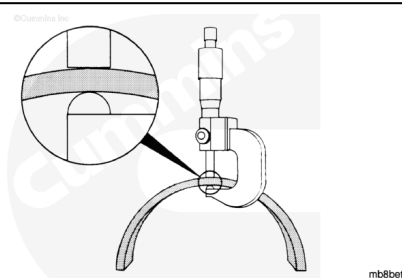
Inspect the bearing seating surface for nicks or burrs.

If nicks or burrs can **not** be removed with a fine crocus cloth, the bearings **must** be replaced.



Measure the rod bearing shell thickness with an outside micrometer that has a ball tip.

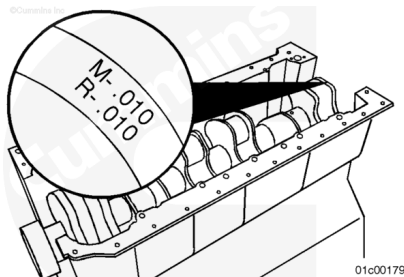
Discard a bearing shell if its thickness is below the minimum specification.



Standard Connecting Rod Bearing Thickness

mm		in
2.780	MIN	0.109
2.792	MAX	0.110

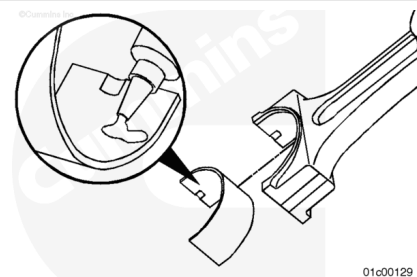
Crankshafts that are ground undersize in the connecting rod or the main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.



Install

⚠ CAUTION ⚠

The connecting rod and bearing shell mating surfaces must be clean and dry when the bearing shells are installed. Used bearings must be installed in their original location or engine damage can occur.

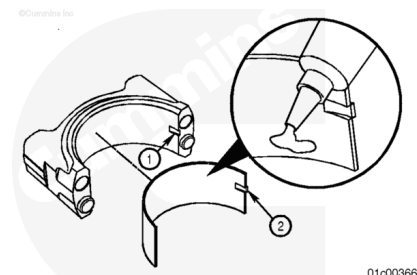


Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the crankshaft journal mating surface of the upper bearing shell.

Install the upper bearing shell into the connecting rod with the tang of the bearing in the slot of the rod.

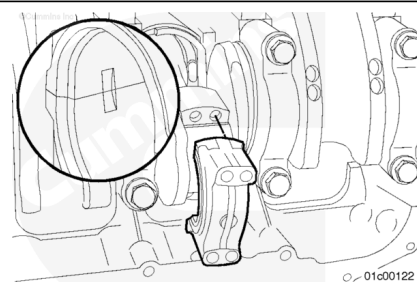
Install the bearing shell into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the bearing shell to crankshaft journal mating surface.



⚠ CAUTION ⚠

Do not damage the fractured split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fractured split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the rod.



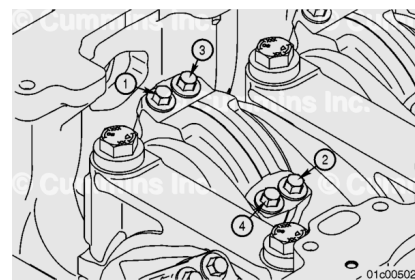
Install the cap onto the connecting rod.

Lubricate the threads and underside of the head of the connecting rod capscrews with clean engine oil.

Thread the capscrews into the connecting rod and hand tighten.

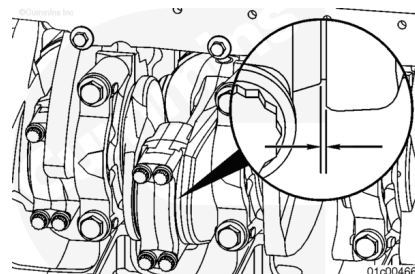
Tighten the capscrews in numerical sequence. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/269/269-001-054-tr.html)



Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.07	MIN	0.0027
0.30	MAX	0.0118



The connecting rod **must** move freely from side to side on the crankshaft journal. If the rod does **not** move freely, remove the rod cap and make sure the bearing shells are the correct size. Check for dirt or damage on the crankshaft and the bearing shells.

Repeat the above steps to install the remaining bearing shells and connecting rod caps.

Finishing Steps

⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after any internal engine repairs to reduce the possibility of internal component damage.



- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-046-tr.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-089-tr.html)

- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/269/269-007-035-tr.html)
- Install the oil transfer connection. Refer to Procedure 007-061 in Section 7.
(/qs3/pubsys2/xml/en/procedures/269/269-007-061-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/269/269-007-025-tr.html)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/269/269-007-037-tr.html)
- Operate the engine until it reaches normal operating temperature and check for leaks.